

We must remember: there
is no easy way.

Ryan Holiday

class starts @ 9:00

Agenda

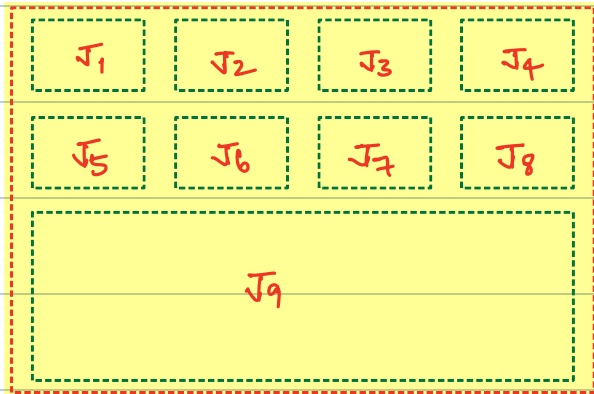
- ① Adaptive Query Execution (AQE)
- ② Advanced configs
- ③ Interview question

SPARK

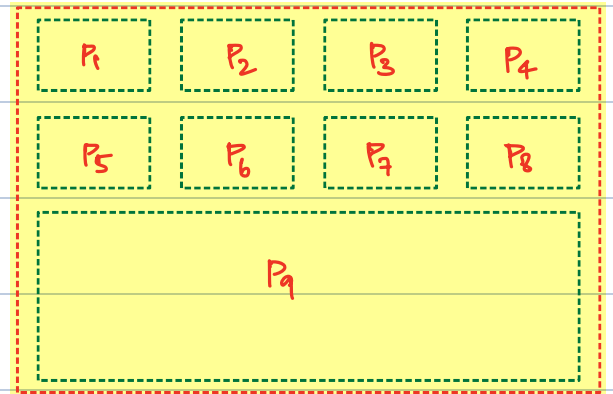
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- ① Introduction to BigQuery
 - ② Load data in BigQuery
 - ③ Optimise

BigQuery

Adaptive Query Execution [AQE]



Project

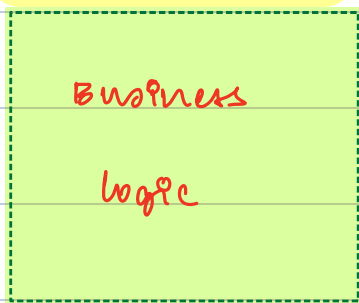


Spark Task

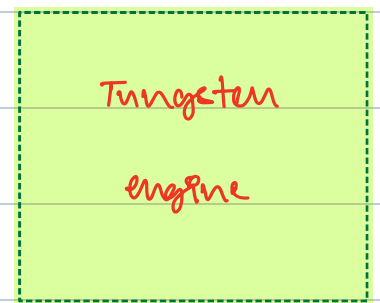
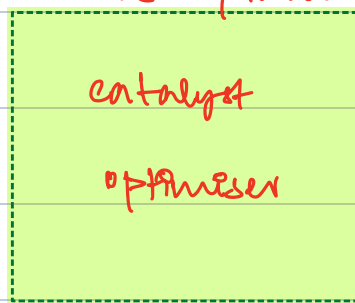
Projects → Jira Tickets

Spark Logic → Tasks

without AQE



→ static plans



shuffle



materialize (local disk)



Pt has necessary metrics.

Main config : spark.sql.adaptive.enabled
3x

Adaptive Query Executor

coalescing shuffle
Partition

switching Join
Strategy

Handle Skewed
Join

Coalescing Shuffle Partition

In a single executor it will merge smaller partitions

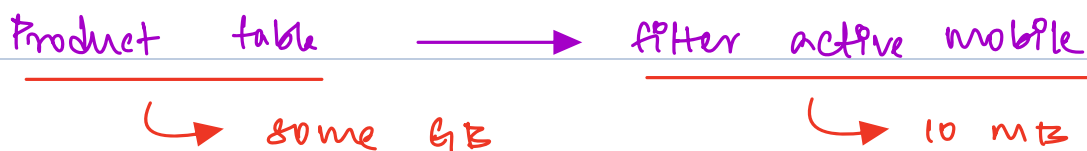
spark.sql.adaptive.coalescePartitions.enabled [true]

spark.sql.adaptive.advisoryPartitionSizeInBytes [64MB]

spark.sql.adaptive.coalescePartitions.minPartitionSize [1MB]

Switching Join Strategy

broadcast vs sort Merge Join



spark.sql.autoBroadcastJoinThreshold [10MB]

Handle skewed Join

spark.sql.adaptive.skewJoin.enabled [true]

spark.sql.adaptive.skewJoin.skewedPartitionThresholdInBytes [256MB]

spark.sql.adaptive.skewJoin.skewedPartitionFactor [5]

If you have predictable data → ARE not very useful

Advanced Config

Spark. sql. dynamic Partition Pruning. enabled $\longrightarrow$ where you

filter based on variable

calculate the filter first and prune unrelated partition from source

Interview Question

How will you handle 100GB of Data in spark?

200 executors, 2GB ram, 4 cores.

How many Partitions? 100GB, 800 partitions

How many cores to handle these partitions? 1 core $\longrightarrow$ 1 par

800 cores

How many cores per executor? [3 to 5 core]

4 cores $\longrightarrow$ 1 executor, 800 cores $\longrightarrow$ 200 ex

How many executors? 200 executor

what should be the RAM per executor?

4 cores, 4 parallel task 128MB

buffer memory x 4 times

$128 * 4 * 4 = 1024 \text{ MB}$ [execution memory]

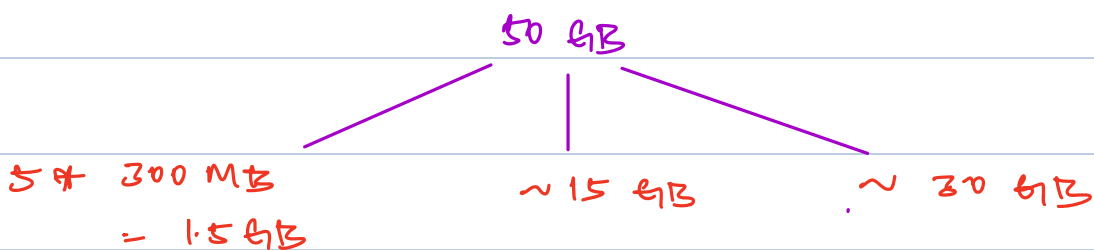
= 2 GB memory

Correct Way

- ① Type of data? $\longrightarrow$ File input in parquet
- ② single or multiple source? $\longrightarrow$ 90GB, 7GB, 3GB, 10MB
- ③ Type of transformations in Business logic? Not clear
- ④ What is the SLA? $\longrightarrow$ 2 Hours, overnight
- ⑤ Cluster configuration? $\longrightarrow$ 2 Drivers, 50 Executors
5 executors to handle other use case

Approach 1:

100 GB $\longrightarrow$ 800 partitions [800 cores]
 $\searrow$ I will make sure overall RAM is at least half of overall size



Reserve memory. user memory. execution memory

executors — 5, 50 GB

per executor — 10 GB of RAM

each of executor — 4 to 5 CPU cores.

Results

① You did not meet your SLA

Sherlock → the next step

① Job took most resources

(a) Data skew

(b) Joins?

(c) which stage causes trouble

① executor memory $\uparrow$

② Data skewedness — salting, cache, repartition
coalesce

③ Is my source read multiple times

cache the solution

④ should $\uparrow$ increase my executor memory.

② You have met your SLA

(a) optimize even further

(b) reduce your resources

continuously monitor & improve
your job

Introduction to BigQuery



Data warehouse by Google



Pizza - Data

Hands - Network

Month - Business processing



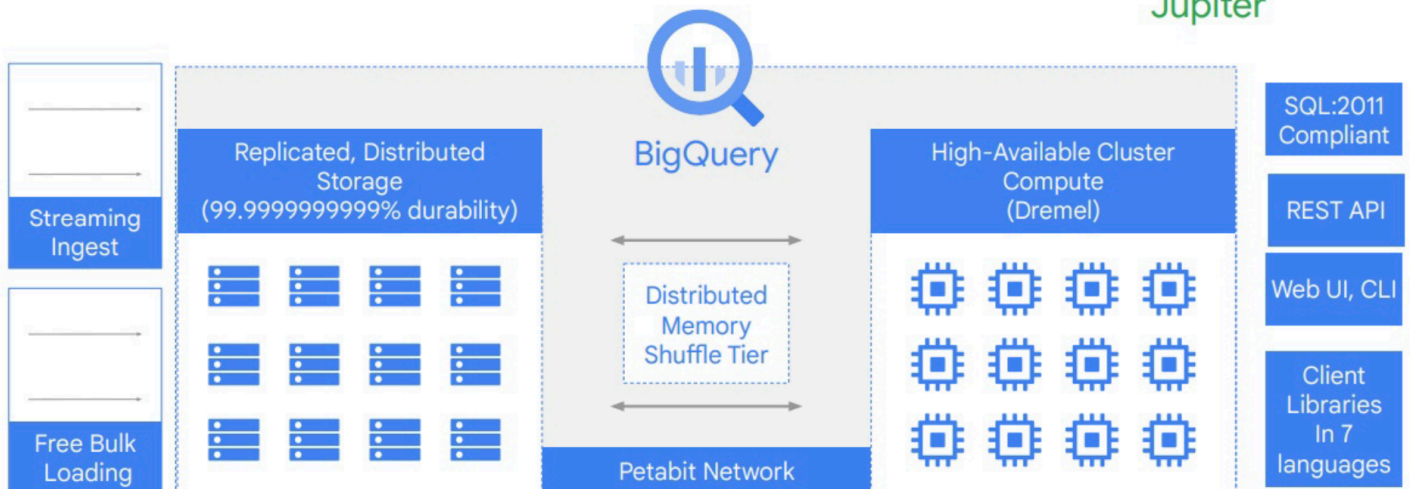
Food - Data

Month - Business logic

very minimal use of

Hands

Colossus
Jupiter



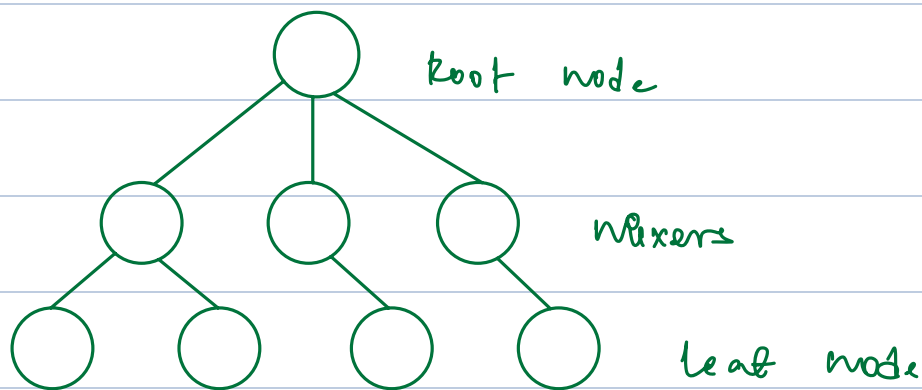
Colossus → Built on top of Google Fik system

HDFS

Tuplex Petabit Network → High Bandwidth network
capable of transferring petabits of data in a sec

Dremel compute

High availability compute only used on Demand
similar to Redshift servers.



Leaf node

Read a partition and perform base level
aggregation

Mixer Aggregate multiple leaf level data.

BigQuery slots

